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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Syntax 404
Selected Domain	Domain 1: The Aiversecurity Future-ready Workforce Domain
Selected SDG Track (1–17)	SDG 8 – Decent Work and Economic Growth
Project / Solution Name	Faraway – Career and Education Compass
Name (Leader)	Low Xin Hui
Email & Phone (Leader)	SWE2509070@xmu.edu.my +60 18-977 3221
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence is changing how work is performed and which skills employers need. However, the pace and effects of these changes remain uncertain across occupations. University students, graduates, and working adults may struggle to understand how their chosen careers could evolve, which existing skills will remain relevant, and what they should learn next. Preparing for only one assumed future can leave them unprepared if AI adoption develops differently.

Career information is also fragmented across occupational databases, labour-market reports, and learning platforms. Users must connect this information to their own abilities, interests, time, and budget. Without accessible guidance, they risk choosing unsuitable learning resources or spending time developing skills that do not address their actual gaps.

The challenge extends to HR teams and universities or TVET institutions. These organisations need to understand how changing skill requirements affect different departments and academic programmes, identify development priorities, and prepare employees or students for several possible futures.

Faraway addresses this uncertainty by helping users compare moderate and rapid AI-development scenarios, examine how assumed skill requirements affect career matches, and translate identified gaps into practical learning plans. These scenarios support preparation and informed decisions; they do not claim to predict exactly how AI or the job market will develop.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

2.2 The Proposed AI Solution

Faraway is an AI-assisted career and education platform for individuals, HR teams, and universities or TVET institutions. It connects skill assessment, career exploration, workforce analysis, and practical learning plans.

Its central feature compares moderate and rapid AI-development scenarios. Users can explore how different assumptions about AI adoption could affect skill requirements, career-match scores, and learning priorities, helping them prepare for uncertainty rather than rely on a single prediction.

Individuals create profiles, assess their skills, and explore related careers with explainable match scores. HR teams organise employees by department, while institutions organise students by academic programme. Career suggestions and trend charts reflect the selected department or programme and its relevant skills.

Application code calculates match scores and skill gaps. Qwen3-4B-Instruct-2507, hosted through RunPod Serverless, generates contextual explanations, SDG connections, and structured learning plans using supplied catalogue information and user preferences. Users can track learning progress and compare their assessed skills against career requirements.

Public evidence includes O*NET occupational information, DOSM labour-market context, and BLS employment projections. Sources retain their dates and geographical limitations. Hypothetical AI scenarios are distinguished from observed statistics and official forecasts. A scheduled workflow updates supported evidence snapshots, while some catalogue updates require review.

The prototype applies GenAI inference, database architecture, and cloud deployment concepts. React, TypeScript, and Vite support the interface; Supabase provides authentication, personal data storage, and backend functions; and Vercel hosts the website. Replit and Codex support development and testing. The interface supports English, Simplified Chinese, and Bahasa Melayu.

Faraway aims to turn uncertainty about AI into understandable career comparisons and practical learning actions.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Faraway uses **React and TypeScript**, with **Vite and Tailwind CSS**, for its responsive interface. **Recharts** supports skill and job-trend visualisations. **Supabase** provides email authentication, PostgreSQL storage for personal profiles and assessments, and Edge Functions for backend AI requests. **Vercel** hosts the website, while **GitHub** maintains the source code.

Our project uses the sponsor platforms Replit and RunPod. Replit supports application development, interface editing, and testing. RunPod Serverless hosts the pretrained **Qwen3-4B-Instruct-2507** model through **vLLM** for AI inference. The model generates contextual explanations, SDG connections, and personalised learning plans. Application code calculates career-match scores, skill gaps, and scenario comparisons.

The AI request flow is:

User interface → Supabase authentication and Edge Functions → RunPod-hosted Qwen model → validated response displayed in Faraway.

Personal records are stored in Supabase PostgreSQL. Organisation workspaces currently use account-scoped browser storage.

Data reliability and transparency are supported through O*NET occupational information, DOSM Malaysian labour-market statistics, and BLS US employment projections. Charts identify their sources, reference periods, and geographical coverage. Skill-importance ratings are distinguished from vacancy counts, and moderate and rapid AI-development scenarios are labelled as Faraway's planning assumptions rather than official forecasts. GitHub Actions schedules supported evidence updates and publishes dated snapshots; some datasets and occupational mappings require reviewed updates.

AI credentials remain in server-side secrets. RunPod workers can scale to zero between requests to manage GPU costs, although cold starts can increase response time. Requests use bounded generation settings, and automatic retries are avoided to reduce duplicate GPU work.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our design process focuses on three user journeys: individual career planning, HR workforce development, and university or TVET student development. We translated these needs into profile assessment, department or programme setup, career comparisons, learning plans, and progress tracking.

We built and deployed an interactive prototype, then refined it through team walkthroughs and functional testing. Testing exposed issues involving CSV imports, new-account isolation, programme-specific career suggestions, and chart interpretation. These findings informed changes to validation, filtering, and interface layout.

The interface supports desktop and mobile layouts and three languages. Numerical scores are calculated by application rules, while AI supports explanations and learning-plan generation.

Deployment uses Vercel, Supabase, and RunPod. Phase 2 priorities include structured user testing, stronger occupational and skill mappings, clearer matching explanations, and verification of evidence updates and access controls. Personal accounts use authenticated storage; organisation records currently remain in account-scoped browser storage.

Our hardest technical risk is maintaining reliable, current evidence that is correctly matched to careers. We address this through source attribution, dated snapshots, validation, and explicit unavailable-data states. We will evaluate user understanding through a defined pilot rather than treating successful deployment as proof of impact.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 4.4 — Relevant skills for employment** By 2030, substantially increase the number of youth and adults with relevant skills for employment, decent jobs, and entrepreneurship. Faraway supports this target by identifying career-related skill gaps and recommending relevant learning pathways under moderate and rapid AI-development scenarios (United Nations, n.d.-a). **SDG 8.5 — Productive employment and decent work** By 2030, achieve full and productive employment and decent work for all, including young people and persons with disabilities, and equal pay for work of equal value. Faraway supports this target through informed career preparation and targeted workforce development as AI changes occupational requirements (United Nations, n.d.-b).

Metric Category	Your Response
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The current baseline is the average percentage of career-related skill gaps correctly identified by participants **without Faraway**. It will be measured through a pre-test using a sample skills profile and a target occupation's documented requirements. **Baseline measurement is pending; no participant results are claimed at this stage.**
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Skill-gap identification accuracy (%):** the percentage of assessed skill requirements correctly classified as either meeting the target or requiring improvement, marked against a prepared answer key.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve a **20% relative improvement in average skill-gap identification accuracy** after using Faraway, compared with the measured pre-test baseline. Equivalent pre-test and post-test tasks will use the same marking method. This is a proposed target, not an achieved result.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Code framework	MIT
TypeScript	Programming language/tooling	Apache 2.0
Vite	Build tooling	MIT
Tailwind CSS	Styling framework	MIT
Recharts	Visualisation library	MIT
Supabase JavaScript client	Code library	MIT
Qwen3-4B-Instruct-2507	Pretrained AI model	Apache 2.0
vLLM	Model-serving software	Apache 2.0
O*NET occupational database	Dataset	CC BY 4.0, subject to stated exceptions; source and adaptations attributed

BLS occupational employment information and projections	Dataset	Public-domain material, subject to stated exceptions; BLS cited
DOSM labour-market publications and data	Dataset	Applicable terms of the specific publication or dataset; source attributed
Meta NLLB-200 distilled 600M	Pretrained translation model used during development	CC BY-NC 4.0

6.3 Employment-Related IP Conflicts

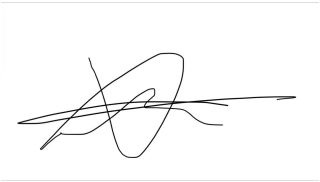
Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date] 24-10-2026